

Primer

Remote sensing for wildfire monitoring: Insights into burned area, emissions, and fire dynamics

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SUMMARY

Remote sensing plays a central role in monitoring wildfires throughout their life cycle, including assessing pre-fire fuel conditions, characterizing active fire locations and emissions, and evaluating post-fire effects on vegetation, air quality, and climate. This primer examines current remote sensing products used in wildfire research, focusing on their application in deriving burned area and emissions data and tracking the dynamic spread of individual fire events. We evaluate the strengths and weaknesses of these products and address key challenges such as generating complete, continuous, and consistent long-term monitoring data. We also explore future opportunities and directions in remote sensing technology for wildfire characterization and management.

INTRODUCTION TO FIRE MONITORING

Fire is a worldwide phenomenon with profound impacts on ecosystem functions, climate change, air quality, and human health and property. Landscape fires, whether ignited by lightning or human activities, vary in frequency, intensity, duration, and spatial extent across different regions. Fire activity also varies seasonally and interannually, often driven by changing weather conditions, vegetation types, and ignition sources. This complex spatial and temporal variability highlights the need for better monitoring strategies to fully track landscape fires.

Historically, fire monitoring has relied on ground-based observations. These observations can be direct visual assessments from watchtowers or more organized monitoring efforts by fire management agencies, including airborne reconnaissance. While these methods provided valuable *in situ* data in specific locations, they were inherently limited in scope, unable to provide the reliable and continuous monitoring required at a continental or global scale.

Remote sensing technology has transformed our ability to observe and analyze fires worldwide. Early space-based fire detection and mapping can be traced back to the 1970s with the launch of the first Landsat satellite. Since then, the evolution of remote sensing technology has produced multiple generations of new instrumentation, each with improved fire detection capabilities. These sensors have been used to collect important data on fire location, area burned, and fire emissions and to generate data streams essential for fire impact analysis.

The continued development of remote sensing, along with advances in data analysis and computational capability, offers promising opportunities for improving our understanding of changing fire dynamics and fire management challenges in a

warmer, more flammable world. Here, we review the current state and future opportunities for remote sensing throughout the fire life cycle, focusing on three applications related to fire monitoring: quantifying burned area (BA), estimating fire emissions (EM), and tracking the dynamic spread of individual fire events (Table 1).

PHYSICAL BASIS FOR REMOTE FIRE MONITORING

Remote sensing is the process of acquiring data about objects or areas without direct physical contact. Dating back to the 19th century, the history of remote sensing starts with the advent of balloon-captured aerographs. Since the mid-twentieth century, various remote sensors have been developed and deployed on airborne and satellite platforms. These sensors serve many purposes, including environmental monitoring, agricultural and urban planning, disaster management, and military services, among others. In particular, the remote sensing instruments on Earth observation (EO) satellites have provided invaluable data for understanding and effectively managing Earth's ecosystems and resources.

Fires significantly impact the Earth's ecosystems, atmosphere, and climate. To facilitate effective fire management and mitigation, people have used remote sensing to monitor the whole life cycle of wildfires. During the pre-fire phase, remote sensing is used to assess fuel loads (amount), fuel condition (e.g., live vs. dead, moisture content), and fire weather to guide predictions of the likelihood and intensity of fire outbreaks. Once a wildfire ignites, remote sensing is critical for real-time information on active fire (AF) locations and intensity. In the post-fire phase, remote sensing assesses changes in burned surfaces (e.g., reflectance, roughness, and temperature) and vegetation (e.g., biomass, structure) to help quantify burn extent and



Table 1. Current satellite remote sensing instruments and their example usages in AF monitoring, BA delineation, and EM estimation

Orbit	Spatial resolution	Instrument (platform)	AF	BA and EM
GEO	coarse	● ABI (GOES-16, -17, and -18) ● FCI (Meteosat 3rd generation) ● SEVIRI (Meteosat 2nd generation) ● AHI (Himawari-8 and -9)	● AF diurnal cycle ● fire spread tracking	● sub-daily distribution of BA and EM
LEO	coarse	● MODIS (Terra, Aqua) ● VIIRS (SNPP, NOAA20, NOAA21) ● SLSTR (Sentinel-3) ● AVHRR (NOAA series, MetOp series) ● VGT (SPOT, Proba-V)	● routine observations of Global AF activity ● NRT FRP-based emissions ● fire events spread tracking	● global BA and EM dataset generation
LEO	medium	● ETM+ (Landsat-7) ● OLI (Landsat-8 and -9) ● TIRS (Landsat-8 and -9) ● MSI (Sentinel-2) ● ASTER (Terra)	● landscape-scale AF analysis ● fire growth assessment	● regional BA and EM datasets ● reference data for global BA datasets
LEO	fine	● Dove (Planet Labs PlanetScope) ● (DigitalGlobe WorldView)	● high-resolution fire line positioning	● post-fire property damage assessment

severity. Remote sensing observations also measure atmospheric composition during and after fires for air quality and pollution analyses.

Remote sensing involves the measurement of electromagnetic radiation that is reflected, emitted, or scattered by a target object or area. Various remote sensing techniques and wavelengths across the electromagnetic spectrum are used for fire monitoring, each serving a specific purpose in the observation and analysis of wildfires (Figure 1).

Thermal imagers, which detect the infrared radiation emitted by fires, are the primary tools for real-time monitoring of AFs. Burning vegetation at temperatures between 500°C and 1,200°C emits significantly more infrared radiation than its surroundings, allowing fires to be identified by their infrared signatures. The mid-wave infrared (MWIR; with wavelengths of 3–5 μm) range is especially effective for fire detection, as it captures the peak in energy release from vegetation fires. To enhance detection, long-wave infrared (LWIR; 8–14 μm) data are often paired with MWIR data, improving contrast and fire identification. Short-wave infrared (SWIR; 1.4–2.3 μm) data are also used for AF detection, particularly during the daytime to avoid confounding effects from reflected sunlight, or for sensors that lack MWIR capabilities.

Measurements of radiances from the solar domain, including visible (VIS; 0.4–0.7 μm), near-infrared (NIR; 0.8–1.4 μm), and SWIR wavelengths, are crucial for assessing the surface and atmospheric changes following a fire. These changes, like vegetation charring, soil exposure, and ash deposition, alter reflectance patterns in these bands. Reflectance in NIR often decreases after a fire, whereas SWIR and red (0.6–0.7 μm) bands typically show increased reflectance. Spectral indices contrasting NIR with SWIR or red bands also help quantify burn severity and identify BAs.

Passive sensors depend on natural energy sources such as fire-emitted thermal radiation or reflected sunlight and are limited by obstacles such as clouds, thick smoke, or vegetation canopies. Active sensors, such as radar (using microwave pulses) and light detection and ranging (LiDAR) (using VIS or NIR lasers),

overcome some of these limits, offering complementary fire monitoring data. They have been employed in both spaceborne and airborne missions to map BAs, assess vegetation structure, and measure smoke plume heights.

The choice of remote sensing platform significantly influences fire monitoring effectiveness. Factors like satellite orbit altitude, inclination, overpass time, and revisit frequency are crucial for wildfire detection performance. While sensors on aircraft and unmanned aerial vehicles (UAVs) offer high spatial resolution and rapid deployment, they have limited spatial coverage and may be restricted by weather conditions. Most remotely sensed data for studying fire characteristics and effects are sourced from EO satellites, operating mainly in low Earth orbit (LEO; below 1000 km), despite limitations in coverage and frequency. LEO satellite imagery varies in resolution, from relatively coarse (hundreds of meters) and medium (tens of meters) to fine (sub-meters to meters), impacting their application and monitoring capabilities. Alternatively, geostationary Earth orbit (GEO; approximately 35,786 km above Earth) satellites provide continuous monitoring of specific regions, although they suffer from lower spatial resolution and reduced effectiveness at high latitudes.

CURRENT REMOTELY SENSED FIRE MONITORING DATA

Remote sensing for BA estimation

Wildfires directly alter the structure and properties of vegetation and soils. By recording reflected or emitted spectral radiances before and after a fire, remote sensing instruments can capture these surface changes to assess the area and severity of the burn (Figure 2). Remotely sensed BA data can be used for a variety of applications, including fire management, fire impact evaluation and remediation, fire driver characterization, and fire risk assessment. They are also often used to estimate greenhouse gas and smoke aerosol emissions from fires, contributing to studies of air quality and climate change.

The predominant method for identifying BAs with remote sensing detects changes at the pixel level in time series of

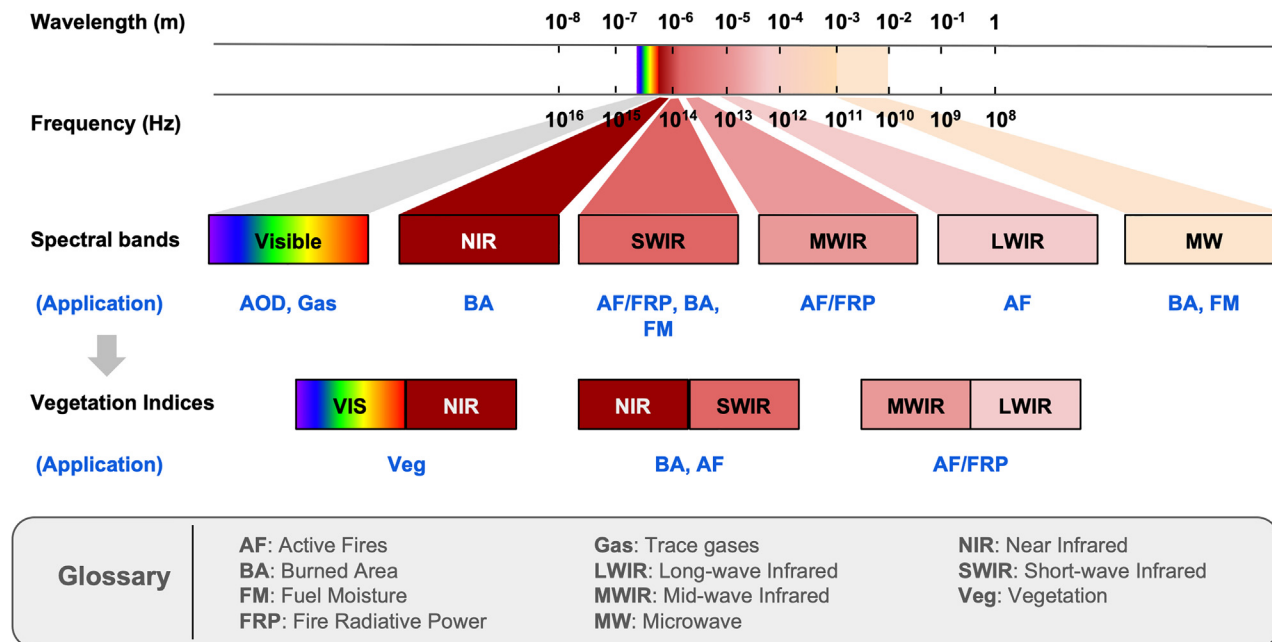


Figure 1. Wavelength-dependent remote sensing applications for fire monitoring

Illustration of the wavelength-dependent applications of remote sensing for monitoring active fires, mapping burned areas, and quantifying fire emissions.

satellite imagery. This method typically involves analyzing radiances or reflectances from specific sensor channels that are sensitive to changes caused by fires. Notably, NIR reflectance usually decreases after fires due to loss of leaf area, and SWIR reflectance increases as vegetation dries and ash accumulates. These changes make NIR and SWIR radiances key components of many current BA algorithms, often combined in vegetation indices such as the normalized burn ratio (NBR) and its derivative, the difference NBR.

Current remotely sensed BA products differ in the spatial and temporal resolution of the imagery and the algorithms used to identify burned and unburned pixels. For global products that typically use daily imagery at coarse resolution from LEO satellites, the standard procedure involves conducting multiple tests on the time series for each pixel to identify the time of sustained change and differentiate burned from unburned pixels. Parameters in these analyses are often empirically determined and locally adjusted. To aid in the selection of training pixels for BAs and to distinguish between pre- and post-burn conditions, AF detections from MWIR and LWIR are sometimes used. Several global BA datasets have been developed using this approach with coarse-resolution remote sensing data. Early global products included ESA's GLOBSCAR, which used data from ATSR aboard the ERS-2 satellite; GBA2000, which used data from VEGETATION aboard the SPOT satellite; and FireCCI41, which used data from MERIS aboard the ENVISAT satellite. Several recent global BA products, including MCD64, FireCCI50, and FireCCI51, are based on the MODIS data from NASA's Terra and Aqua satellites.

Measurements from medium-resolution sensors, such as ETM+ and OLI on Landsat, MSI on Sentinel-2, and ASTER on Terra, are widely used for regional BA mapping and training/validation of global products. These regional products employ BA

algorithms similar to the global products, using time series of single- or multi-band data. Unlike physics-based models, machine learning techniques are often used for pixel-level burn probability and classification, taking advantage of recent advances in open data access and computing power. Examples include USGS BA in the US, MapBiomas in Brazil, NBAC in Canada, TERN in Queensland of Australia, FireCCI SFD11 in sub-Saharan Africa, and Greenpeace in Russia—products that primarily use Landsat or Sentinel-2 data. Although machine learning approaches offer speed and require less empirical knowledge, their outputs lack physical interpretability and may be highly sensitive to noise and outliers.

The calibration and validation of BA data involve comparisons with independent, higher-resolution reference imagery. The accuracy of global, coarse-resolution BA products is often assessed by calculating commission (false positives) and omission (missed fires) errors against reference areas from carefully selected Landsat or Sentinel-2 scenes representing diverse locations and times. Many reference datasets are quality checked through visual inspection and compiled into the Burned Area Reference Database. Commercial satellites and aerial photos may offer some reference data for validating medium-resolution products, but their coverage is limited, and data acquisition can be costly. Currently, medium-resolution reference data or expert interpretation of fine-resolution imagery is commonly used to assess the accuracy of Landsat- and Sentinel-2-based BA products.

To better examine the long-term effects of fires on ecosystems, air quality, and climate, consistent global BA datasets have been created by combining information from multiple satellite platforms. The Global Fire Emissions Database BA (GFED BA) exemplifies this approach, using the MODIS MCD64 BA product as its primary input. AF data from the ATSR and VIRS

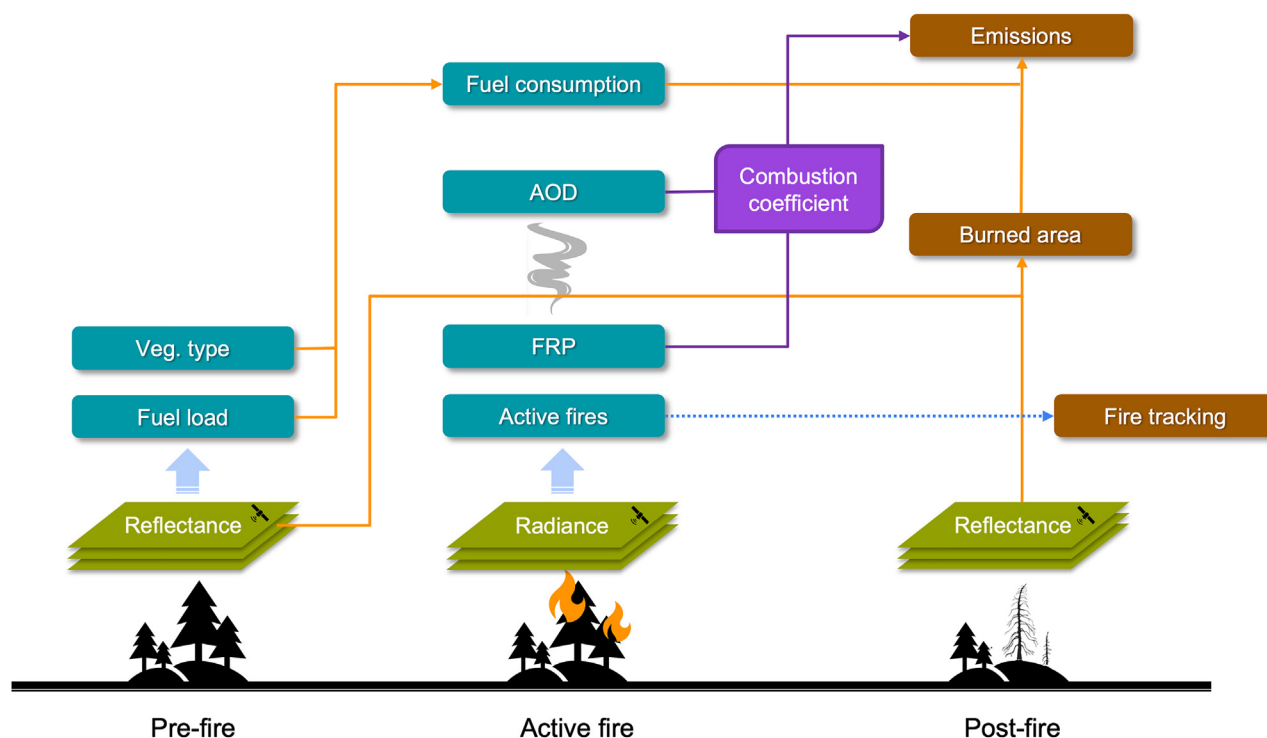


Figure 2. Illustration of the remote sensing workflow for fire monitoring, detailing the pre-fire, active fire, and post-fire stages

The diagram shows how reflectance and radiance data from remote sensing are used to estimate emissions, map area burned, and track individual fire events.

sensors are used to extend the data record prior to the MODIS era. In its latest iteration (GFED5), MODIS AF observations and medium-resolution reference data help estimate areas affected by smaller or cooler fires (e.g., smoldering, understory, or sub-surface fires and low-intensity grassland and brush fires) that may be missed by MCD64. In addition, to accurately capture cropland fires, which are typically smaller and quickly obscured by planting, a special BA reference dataset was developed using a mixture of medium- and fine-resolution (PlanetScope) imagery. Finally, to generate a higher temporal resolution emissions dataset, the monthly BA data from the GFED were distributed to sub-daily intervals using AF detections and diurnal cycles from coarse but frequently recorded geostationary satellite data (GOES-R, HIMAWARI, Meteosat).

Remote sensing for estimating emissions from fires

Early fire emission estimates mainly relied on national or regional fire statistics. The introduction of remotely sensed fire data has improved the accuracy of regional and global emissions estimates, including the total amounts and spatiotemporal patterns of trace gasses and aerosols from landscape fires. There are three primary methods for using remotely sensed data to estimate fire emissions.

The first method, known as the process-based approach, involves a detailed analysis of the fire's burning process. This includes calculating the available fuel, area burned, fuel consumed, and emissions produced. The formula for estimating gas and aerosol species emitted from fires (E) follows several sequential steps: $E_i = A \times B \times C \times EF_i$, where A is the area burned; B is the biomass available per area; C is combustion

completeness, representing the fraction of biomass actually consumed; and EF_i is the emission factor for species i . This approach incorporates remote sensing data for estimating area burned (A), discussed in the previous section. Estimates of available biomass (B) and combustion completeness (C) vary from simple averages by biome type to complex dynamic vegetation models that track carbon stocks and fluxes and often depend on remotely sensed data such as vegetation indices and fuel moisture. This approach has produced most of the current fire emissions datasets, including GFED, FLAMBE, GICC, WFEIS, MFLEI, and FINN, reflecting the broad application of this strategy.

The second approach is physically based on the linear correlation between remotely sensed fire radiative energy and biomass consumed in fires, independent of vegetation type. In this method, the emission rate of gasses and particles is directly derived from the fire radiative power (FRP), with total emissions calculated by temporal integration: $E_i = \alpha \times \text{FRE} \times EF_i = \alpha \times \int \text{FRP} \times EF_i$. The combustion coefficient α is determined by experiments or calibration from other existing emissions datasets. For example, MODIS data on FRP and fire location have supported the creation of global biomass burning emission products such as QFED and GFAS. Some studies have inversely calculated species-specific combustion coefficients using remotely sensed data, such as correlating FRP with aerosol optical depth measurements from MODIS to produce the FEER product. LEO satellites typically capture small fires more accurately, while GEO satellites provide more consistent measurement intervals, allowing for better temporal integrations of FRP and emissions in regional products such as RAVE. The FRP approach provides near-real-time (NRT) emission estimates by bypassing the

detailed BA and fuel analysis; however, its accuracy may not be as good as the process-based method due to limited coverage from current satellites.

In the third approach, satellite instruments measure atmospheric concentrations of fire-emitted pollutants (CO, NO_x, CO₂) and integrate these with atmospheric inversion models to refine fire emission estimates, known as the “inversion-based approach.” Although model resolution and uncertainties can impact the quality of these indirectly derived emission datasets, they provide valuable insights to enhance process-based emission estimates through top-down constraints.

Heat release from fires results in buoyant gas and aerosol dispersion that is not confined to the surface. Satellite EO datasets have also been used to provide information on the height distribution of fire plumes. For example, multi-angle stereo imagery from the MISR instrument has been used to determine smoke plume heights. Aerosol profiles measured by satellite-based LiDAR instruments such as CALIPSO can also be used to evaluate smoke injection height. However, it remains a challenge to separate the rapid, strong updraft due to thermal energy from the subsequent atmospheric loft, given the sparse availability of vertical profile measurements.

Remote sensing for AFs and spread tracking

AFs are fires that are currently burning and producing detectable heat, light, and smoke. Remote sensing techniques have been used to determine the location and energy release (FRP) of AFs. Unlike BA data, whose algorithms often require the inspection of post-fire imagery, remotely sensed AF and FRP data can be obtained in near-real time. This makes AF information particularly suitable for fire detection and early warning, fire behavior analysis and spread prediction, and air quality monitoring. As described in previous sections, AF detection is also used in some BA and fire emission algorithms. For example, AF data are often used to help determine the BA training data, and some global fire emission datasets, especially those used for NRT applications, often rely on direct measurements of radiative energy from the AF pixels.

Most sensors use radiance measurements from multiple infrared bands to identify AF locations. The radiance difference between MWIR and LWIR is often used for coarse-resolution sensors such as MODIS and VIIRS due to the high contrast in these wavelengths between pixels with and without fire. Medium-resolution sensors such as Landsat and Sentinel-2 lack an MWIR channel and therefore use SWIR and NIR bands for AF detection in cloud-free imagery (which is possible due in part to their higher spatial resolution). To reduce omission and commission errors, the operational AF algorithms for these sensors often involve a series of testing procedures, such as initial screening, threshold testing, contextual testing, and rejection testing, using multiple radiances and vegetation indices. The algorithms frequently differ between day and night, given the added complexities for AF detection from reflected sunlight.

AFs and BAs detected by remote sensing typically appear as discrete pixel-level data. Recent studies have aimed to link these data points by aggregating nearby fire pixels into objects to enable dynamic tracking of individual fire events over time. Effective fire object tracking relies on several key factors: timely data availability, accurate geolocation of fire pixels, and short

detection return times to minimize gaps in tracking fire spread. Commonly used data sources include BA from MODIS (e.g., Global Fire Atlas) and AF detections from VIIRS (e.g., FEDS) and GOES-R (e.g., GOFER). These fire object-based data are valuable for training fire spread models, analyzing fire risk and history, and estimating carbon emissions from specific fires.

LIMITATIONS AND CHALLENGES

While remote sensing is a crucial tool for fire monitoring, various data and algorithm limitations affect the accuracy and reliability of remotely sensed estimates of BAs, fire emissions, and fire spread.

One primary issue is that most current EO sensors were not specifically designed and launched for fire monitoring, and some limitations of these general-purpose sensors contribute to errors of commission and omission in AF and BA products. AF detections often suffer from false alarms due to bright surfaces reflecting sunlight that cannot be completely screened out using available image bands. BA algorithms can misinterpret land-use changes as fire activity, while small, cool, understory, and sub-surface fires are frequently omitted. Global BA products from coarse-resolution sensors often underestimate fire activity compared to higher-resolution imagers or atmospheric inversion methods.

Infrequent spatiotemporal coverage from LEO sensors results in data gaps and missed fire events, particularly in regions with frequent cloud cover or management fire activity between overpass times. Long revisit intervals of LEO satellites can hinder tracking fast-moving fires and estimating near-real-time emissions, as fires can change significantly between satellite passes. Rapid vegetation regrowth in ecosystems like grasslands and croplands can obscure recent fire evidence, complicating BA estimation.

Data integration poses additional challenges. Developing algorithms that perform well globally is difficult due to regional differences in vegetation, fire behavior, and climate. By contrast, algorithms tailored to specific regions may not be applicable elsewhere. Ensuring temporally consistent datasets for long-term trend analysis is critical, whether using global or regional algorithms, as changes in satellite sensors, data processing, and observing conditions can introduce inconsistencies or mask subtle but important changes in regional fire activity.

Validating remote sensing fire data is essential but challenging due to the need for coincident, accurate ground-truth data. This often requires access to costly commercial high-resolution imagery or datasets from limited local experiments or airborne measurements. Scaling up ground observations or limited reference datasets to validate satellite-derived estimates over large areas remains a complex task.

Future directions and opportunities

Addressing the challenges outlined in the previous section will require continued advances in remote sensing technology, algorithm development, validation methods, and integration with other data sources and ground-based observations. These efforts also present opportunities for enhancing fire monitoring through remote sensing, improvements that are urgently needed to understand, anticipate, and manage more extreme fires due to climate change.

One significant challenge is the end of life for the MODIS instruments on Terra and Aqua, which have provided many important global fire datasets for more than 20 years. The VIIRS sensors on multiple NASA and NOAA platforms partially fill this looming gap in global coverage, but continued investment in next-generation sensors and platforms is essential for uninterrupted data flows and long-term consistency in global fire monitoring. The CSA WildFireSat mission, planned as a constellation of smallsat platforms, is one example of the type of technology advance in remote sensing that will be needed to provide more frequent coverage for fire detection and tracking.

The growing number of medium- and high-resolution satellites, such as Sentinel-2a and Sentinel-2b, Landsat 8 and Landsat 9, and commercial platforms like Planet Labs PlanetScope and DigitalGlobe WorldView, provide detailed imagery for detecting and mapping BAs with unprecedented accuracy. With more remotely sensed imagery from these instruments, it is becoming possible to derive global, medium-resolution BA datasets, allowing more accurate assessments of fire impacts on various ecosystems and improving the quality of fire emission inventories. However, coordinating data collection, analysis, and synthesis across multiple civil space and commercial platforms is a substantial hurdle for routine BA mapping on a global basis.

The development of remote hyperspectral imaging sensors is another promising direction for fire monitoring. Hyperspectral sensors capture a wide range of wavelengths, offering detailed information about the composition and condition of vegetation (fuels), soil, and other materials. This spectral detail can enhance the detection of fire-affected areas, improve understanding of fire behavior and fuel characteristics, and contribute to more accurate modeling of fire emissions and impacts to ecosystems and communities.

In addition to new remote sensing technologies, developing algorithms to fully exploit existing sensors and data is crucial. For example, algorithms that process high-resolution imagery and integrate data from multiple sources will improve the accuracy of BA and fire emission estimates. Developing new fire algorithms for older sensors like AVHRR, which have monitored the Earth's surface for decades, can be particularly useful for assessing long-term trends in fire occurrence and their relationship to climate and human activities. Multi-sensor approaches, combining data from thermal, optical, and radar sensors, can mitigate limitations such as cloud cover obscuration or saturation issues, offering more comprehensive and reliable fire monitoring.

Creating a global fire emissions database that includes event-based emissions from large fires is another important future direction. Such a database would facilitate global tracking of fire emissions, assessments of fire impacts on climate and air quality, and understanding the broader environmental and societal impacts of fires.

In summary, the future of remote sensing for fire monitoring is promising, with numerous opportunities for technological advancement and interdisciplinary collaboration. Continued efforts to improve data continuity, utilize high-resolution assets, develop new sensors and algorithms, integrate multiple data sources, and create comprehensive global databases will enhance our ability to monitor, understand, and manage fires in a changing climate.

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DECLARATION OF INTERESTS

The authors declare no competing interests.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used ChatGPT in order to improve language and readability. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

RECOMMENDED READING

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